The Æther-flow Interpretation of Relativity

Higher-Derivative Quartic Branch Unit-Lapse Trace-Monopole Compatibility Test

Abstract

The fixed-completion unit-lapse spatial-harmonic route has already passed its first obstruction note: the straightforward unweighted decaying Sobolev (K, χ, β) package fails whenever the longitudinal source monopole

$$\mathfrak{m}_\chi(t) = \int_{\mathbb{R}^3} \big(K - \mathcal{N}_\chi\big)(t, x)\, dx$$

is nonzero. The smallest possible repair is therefore narrower than a weighted-space redesign. One should first ask whether the same fixed-completion formulation already forces $\mathfrak{m}_\chi(0) = 0$ from its existing constraints and gauge conditions and, if so, whether the unit-lapse evolution propagates that cancellation.

This note performs exactly that test. The verdict is negative at the present recorded scope. First, the existing unit-lapse compatible-data conditions do not presently furnish a derivation of $\mathfrak{m}_\chi(0) = 0$: the Einstein Hamiltonian and momentum constraints, the unit-lapse spatial-harmonic gauge conditions, and decaying auxiliary solvability do not by themselves yield a global identity cancelling the trace monopole. Second, even if one imposes $\mathfrak{m}_\chi(0) = 0$ as an extra initial condition, the current unit-lapse evolution gives only a formal identity for $\frac{d}{dt}\mathfrak{m}_\chi$, and that identity is not closed by any recorded cancellation law. In particular, no theorem-level relation presently ties the integral of the trace evolution to the material derivative of $\mathcal{N}_\chi$.

The conclusion is correspondingly narrow. The propagated trace-monopole cancellation route does not yet repair the unweighted (K, χ, β) architecture on the fixed completion $S_{\min,4}$. A vanishing trace monopole would be a new additional compatibility condition, and its persistence is not currently proved. Therefore the next smallest honest repair is the longitudinal auxiliary renormalization

$$\chi = \chi_{\mathrm{Coul}} + \chi_{\mathrm{reg}},$$

while the broader weighted or homogeneous reformulation should still be deferred until after that smaller fixed-completion route is tested.

1. Introduction

The higher-derivative quartic branch has now reached a sharper fixed-completion decision point. The gauge-first unit-lapse spatial-harmonic reduction removed the old maximal-gauge lapse obstruction at the equation level, but the subsequent trace/auxiliary note showed that the direct unweighted decaying Sobolev replay of the old local and coercive architecture still fails: the longitudinal ordered-flow solve

$$\kappa_\theta \Delta_\gamma \chi = -K + \mathcal{N}_\chi \tag{1}$$

generically produces a Coulomb $1/|x|$ tail whenever the source monopole is nonzero.

That failure left three smaller repair routes on the table:

1. a propagated trace-monopole cancellation
2. a renormalized longitudinal auxiliary split
3. a weighted or homogeneous function-space reformulation.

The present note tests only the first of these because it is the narrowest structural change. If the existing constraints, gauge conditions, and unit-lapse evolution already force

$$\mathfrak{m}_\chi(t) = 0,$$

then the old unweighted theorem architecture could be replayed with the least possible alteration. If they do not, the next honest step is the Coulomb split rather than a larger analytical redesign.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that boundary. It does not claim that no deeper hidden identity could ever be found. It records only that, on the current fixed completion $S_{\min,4}$ and with the unit-lapse reduced system presently on record, the propagated trace-monopole cancellation route does not yet close.

2. The Repair to Be Tested

Work in the same asymptotically flat small-data setting and the same unit-lapse spatial-harmonic gauge

$$N \equiv 1, \qquad V^i(\gamma) = 0, \tag{2}$$

with reduced variables

$$(\gamma_{ij}, \Sigma_{ij}, K; \beta^i; \sigma, \pi_\sigma; \psi, \pi_\psi) \tag{3}$$

and slice-wise auxiliary unknowns (Q^I, χ, W_i^T). As in the previous unit-lapse notes, define

$$F_\chi \equiv K - \mathcal{N}_\chi[\gamma, \Sigma, K, \beta, \sigma, \pi_\sigma, Q, \chi, W^T, \psi, \pi_\psi]. \tag{4}$$

Then (1) becomes

$$\kappa_\theta \Delta_\gamma \chi = -F_\chi. \tag{5}$$

Definition 1 (Trace monopole)**.** *Assume $F_\chi(t,\cdot) \in L^1(\mathbb{R}^3) \cap H^{N-2}(\mathbb{R}^3)$ on a slice. The corresponding*

trace monopole *is*

$$\mathfrak{m}_\chi(t) \equiv \int_{\mathbb{R}^3} F_\chi(t,x)\,dx = \int_{\mathbb{R}^3} (K - \mathcal{N}_\chi)(t,x)\,dx, \tag{6}$$

where dx denotes the asymptotically Cartesian coordinate volume measure used throughout the unit-lapse reduction.

Definition 2 (Propagated trace-monopole cancellation test)**.** *The* propagated trace-monopole cancellation test *asks whether the current unit-lapse formulation already provides both of the following:*

1. *a derivation of $\mathfrak{m}_\chi(0) = 0$ from the existing compatible-data requirements*
2. *a propagation law implying that $\mathfrak{m}_\chi(0) = 0$ forces $\mathfrak{m}_\chi(t) = 0$ for later times in the same regularity class.*

Remark 1. *Definition 2 is deliberately narrower than any weighted-space reformulation. If it closed, then the already-recorded unweighted local and coercive machinery would be the natural next theorem architecture to replay on the repaired package.*

3. Initial-Slice Compatibility

The first question is whether the existing compatible-data conditions already force the monopole to vanish on the initial slice.

Proposition 1 (Decaying auxiliary solvability does not by itself force zero monopole)**.** *Fix a slice in the flat-background small-data regime and assume $F_\chi \in L^1(\mathbb{R}^3) \cap H^{N-2}(\mathbb{R}^3)$, $N \geq 2$. Let χ be the decaying solution of* (5)*. Then*

$$\chi(x) = -\frac{1}{4\pi\kappa_\theta} \int_{\mathbb{R}^3} \frac{F_\chi(y)}{|x-y|}\,dy + \mathcal{R}_\gamma(x), \tag{7}$$

where $\mathcal{R}_\gamma(x) = o(|x|^{-1})$ in the small-data regime. In particular,

$$\chi(x) = -\frac{\mathfrak{m}_\chi}{4\pi\kappa_\theta |x|} + o(|x|^{-1}) \qquad \text{as } |x| \to \infty. \tag{8}$$

Proof. In the small-data regime the spatial metric γ is a perturbation of the Euclidean metric, so standard perturbative elliptic potential theory represents χ by the Euclidean Newton kernel plus a remainder whose contribution is $o(|x|^{-1})$ at spatial infinity. Expanding

$$\frac{1}{|x-y|} = \frac{1}{|x|} + \mathcal{O}\Big(\frac{|y|}{|x|^2}\Big)$$

for large $|x|$ and using $F_\chi \in L^1$ yields

$$\int_{\mathbb{R}^3} \frac{F_\chi(y)}{|x-y|}\,dy = \frac{1}{|x|} \int_{\mathbb{R}^3} F_\chi(y)\,dy + \mathcal{O}(|x|^{-2}).$$

Substituting (6) gives (8). □

Proposition 2 (The existing compatible-data conditions do not presently derive $\mathfrak{m}_\chi(0) = 0$)**.** *At the present recorded scope, the unit-lapse compatible-data requirements do not furnish a derivation of the identity*

$$\mathfrak{m}_\chi(0) = 0. \tag{9}$$

More precisely, the currently displayed data conditions consist of:

1. *the Einstein Hamiltonian and momentum constraints*
2. *the unit-lapse spatial-harmonic gauge conditions*
3. *asymptotic flatness and decaying auxiliary solvability.*

Those conditions do not presently close a proof of (9)*.*

Proof. The Einstein Hamiltonian and momentum constraints are local slice equations for $R[\gamma]$ and $D^j(K_{ij} - \gamma_{ij}K)$. The spatial-harmonic gauge is likewise a local condition on the coordinates and the shift equation used to preserve that gauge. None of the displayed initial conditions contains the global functional $\int_{\mathbb{R}^3}(K - \mathcal{N}_\chi)\,dx$.

Moreover, item (3) is strictly weaker than (9). By Proposition 1, the longitudinal elliptic equation (5) admits decaying solutions whose leading asymptotic behavior is Coulombic whenever $\mathfrak{m}_\chi \neq 0$. Thus the already-recorded requirement that the auxiliary equation admit a decaying solution does not itself force the stronger cancellation (9).

Therefore a vanishing initial trace monopole is not currently derived from the existing compatibility package. At present it would have to be added as a new extra condition rather than read off from the old one. □

Corollary 1 (A zero initial monopole would be a new compatibility condition)**.** *If one wishes to enforce $\mathfrak{m}_\chi(0) = 0$ on the fixed-completion unit-lapse branch, that requirement is presently an additional global compatibility condition rather than a consequence of the existing unit-lapse constraints and gauge conditions.*

Proof. This is exactly Proposition 2. □

4. Formal Monopole Evolution Identity

The next question is whether the unit-lapse evolution nevertheless propagates the cancellation if it is imposed at $t = 0$.

Proposition 3 (Formal coordinate-volume evolution law for the trace monopole)**.** *Assume a sufficiently regular asymptotically flat solution of the unit-lapse reduced system on $[0, T]$ such that $F_\chi \in C^1_t L^1_x$ and $\beta \in C^0_t W^{1,\infty}_x$ with the boundary fluxes decaying sufficiently fast at spatial infinity. Then*

$$\frac{d}{dt}\mathfrak{m}_\chi(t) = \int_{\mathbb{R}^3} \Big((\partial_t - \mathcal{L}_\beta)K - (\partial_t - \mathcal{L}_\beta)\mathcal{N}_\chi - (\partial_i \beta^i)\, F_\chi \Big)\, dx \tag{10}$$

$$= \int_{\mathbb{R}^3} \Big(R[\gamma] + \Sigma_{ij}\Sigma^{ij} + \frac{1}{3}K^2 + \Theta_K - (\partial_t - \mathcal{L}_\beta)\mathcal{N}_\chi - (\partial_i \beta^i)\, F_\chi \Big)\, dx. \tag{11}$$

Proof. Differentiate (6) under the integral sign:

$$\frac{d}{dt}\mathfrak{m}_\chi(t) = \int_{\mathbb{R}^3} \partial_t F_\chi \, dx.$$

For any scalar f,

$$\partial_t f = (\partial_t - \mathcal{L}_\beta) f + \beta^i \partial_i f.$$

Integrating the transport term by parts and using the assumed decay gives

$$\int_{\mathbb{R}^3} \beta^i \partial_i f \, dx = -\int_{\mathbb{R}^3} (\partial_i \beta^i) f \, dx.$$

Applying this to $f = F_\chi$ yields (10). The unit-lapse trace evolution equation

$$(\partial_t - \mathcal{L}_\beta) K = R[\gamma] + \Sigma_{ij}\Sigma^{ij} + \frac{1}{3}K^2 + \Theta_K \tag{12}$$

then gives (11). □

The Hamiltonian constraint rewrites the curvature term in (11).

Proposition 4 (Constraint-reduced form of the monopole evolution identity)**.** *Using the Hamiltonian constraint*

$$R[\gamma] + K^2 - K_{ij}K^{ij} = 16\pi G_N(\rho^{\text{matter}} + \rho_\sigma) \tag{13}$$

and the decomposition

$$K_{ij} = \Sigma_{ij} + \frac{1}{3}\gamma_{ij}K, \qquad K_{ij}K^{ij} = \Sigma_{ij}\Sigma^{ij} + \frac{1}{3}K^2, \tag{14}$$

equation (11) *becomes*

$$\begin{aligned}\frac{d}{dt}\mathfrak{m}_\chi(t) = &\int_{\mathbb{R}^3} \Big(16\pi G_N(\rho^{\text{matter}} + \rho_\sigma) + 2\Sigma_{ij}\Sigma^{ij} - \frac{1}{3}K^2 + \Theta_K\Big)\, dx \\ &- \int_{\mathbb{R}^3} \Big((\partial_t - \mathcal{L}_\beta)\mathcal{N}_\chi + (\partial_i\beta^i)\, F_\chi\Big)\, dx.\end{aligned} \tag{15}$$

Proof. From (13) and (14),

$$R[\gamma] = 16\pi G_N(\rho^{\text{matter}} + \rho_\sigma) + \Sigma_{ij}\Sigma^{ij} - \frac{2}{3}K^2.$$

Insert this into (11). □

Remark 2. *Equation* (15) *is the precise formal quantity one would need to reduce to zero in order to propagate the cancellation* $\mathfrak{m}_\chi(t) = 0$ *in the coordinate-volume sense used by* (6)*.*

5. Why the Propagation Test Does Not Close

The central issue is now clear: does the current architecture prove that the right-hand side of (15) vanishes when $\mathfrak{m}_\chi = 0$?

Proposition 5 (The recorded unit-lapse equations do not presently close $\frac{d}{dt}\mathfrak{m}_\chi = 0$)**.** *At the present recorded scope, the unit-lapse formulation does not provide a closed proof that*

$$\mathfrak{m}_\chi(0) = 0 \qquad \Longrightarrow \qquad \mathfrak{m}_\chi(t) = 0 \quad \textit{for } t > 0. \tag{16}$$

More precisely, the current notes do not supply any identity reducing the right-hand side of (15) *to zero from the existing constraints and gauge conditions alone.*

Proof. Proposition 4 shows that the formal derivative of $\mathfrak{m}_\chi$ depends on two kinds of terms.

First, there are explicit Einstein and matter densities:

$$16\pi G_N(\rho^{\text{matter}} + \rho_\sigma) + 2\Sigma_{ij}\Sigma^{ij} - \frac{1}{3}K^2 + \Theta_K.$$

These are not themselves constrained to integrate to zero by any currently displayed unit-lapse identity.

Second, there are the terms

$$(\partial_t - \mathcal{L}_\beta)\mathcal{N}_\chi, \qquad (\partial_i\beta^i)\, F_\chi,$$

which involve the full lower-order auxiliary and shift structure. At the present manuscript scope, $\mathcal{N}_\chi$ has been recorded only schematically as a lower-order nonlinear functional vanishing on the flat background. No theorem-level formula has been derived showing that its material derivative cancels the integral of the Einstein and matter densities above, and no gauge identity has been recorded that removes the shift-compressibility term in the coordinate-volume functional (6).

Therefore the present architecture does not close a proof of (16). It leaves the desired propagation law unproved. □

Corollary 2 (The propagated trace-monopole cancellation repair does not presently close)**.** *Even if one imposes* $\mathfrak{m}_\chi(0) = 0$ *as an additional initial compatibility condition, the current unit-lapse equations do not yet justify using*

$$\mathfrak{m}_\chi(t) = 0$$

as a propagated slice-wise input to the unweighted (K, χ, β) *closure argument.*

Proof. This is precisely Proposition 5. □

6. Verdict for the Unweighted Architecture

The point of the present note is not simply that a proof is incomplete. It is that the first proposed repair has now been tested at the right theorem level.

Theorem 1 (Verdict of the propagated trace-monopole cancellation test)**.** *Assume the fixed-completion unit-lapse spatial-harmonic reduced system for* $S_{\min,4}$ *in the same asymptotically flat small-data regime used by the previous unit-lapse obstruction note. Then, at the present recorded scope:*

1. *the existing unit-lapse compatible-data conditions do not presently derive* $\mathfrak{m}_\chi(0) = 0$
2. *imposing* $\mathfrak{m}_\chi(0) = 0$ *would amount to adding a new global compatibility condition*

3. *the recorded unit-lapse equations do not presently close a propagation proof for that added condition*

4. *therefore the propagated trace-monopole cancellation route does not yet repair the old unweighted local well-posedness, continuation, or coercive architecture for the* (K, χ, β) *package.*

Proof. Items (1) and (2) are Proposition 2 and Corollary 1. Item (3) is Proposition 5. Item (4) follows because the previous unweighted architecture requires a slice-wise H^N control of χ throughout the evolution interval, not merely at one initial time. Since the cancellation condition is neither derived from the old data nor proved to propagate, the repaired unweighted closure package is not currently available. □

Corollary 3 (Immediate next repair at the fixed completion)**.** *After Theorem 1, the next smallest honest fixed-completion repair is the longitudinal auxiliary renormalization*

$$\chi = \chi_{\mathrm{Coul}} + \chi_{\mathrm{reg}}, \tag{17}$$

followed by a test of whether the regular part χ_{reg} *closes in the old unweighted energy. The broader weighted or homogeneous reformulation should come only after that narrower route fails.*

Proof. Theorem 1 shows that the first and narrowest repair—propagated monopole cancellation within the existing architecture—does not presently close. The Coulomb split is therefore the next least invasive modification of the low-frequency longitudinal treatment while keeping the fixed completion $S_{\mathrm{min},4}$, the same gauge, and the same basic theorem package as close as possible to the earlier one. □

7. Conclusion

The previous unit-lapse trace/auxiliary obstruction note reduced the next task to three small repair routes. The present note tests the narrowest of them.

Its verdict is disciplined and specific. The current unit-lapse compatible-data package does not presently imply $\mathfrak{m}_\chi(0) = 0$, and the current unit-lapse evolution does not presently prove that such a cancellation would propagate even if it were imposed. Thus the obstruction is not yet removed by a propagated trace-monopole cancellation inside the recorded formulation.

That failure is still narrower than a weighted-space redesign. It says only that the first repair does not yet close. The next honest step is therefore the Coulomb decomposition $\chi = \chi_{\mathrm{Coul}} + \chi_{\mathrm{reg}}$ and a direct test of whether the regular remainder closes in the old unweighted energy. Only after that smaller fixed-completion repair fails should the project move to the larger weighted or homogeneous function-space reformulation.